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REVIEW ARTICLE



CE Neural correlates of socioeconomic status in early childhood: A systematic review of the literature

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ABSTRACT

It is now established that socioeconomic variables are associated with cognitive, academic achievement, and psychiatric outcomes. Recent years have shown the advance in our understanding of how socioeconomic status (SES) relates to brain development in the first years of life (ages 0–5 years). However, it remains unknown which neural structures and functions are most sensitive to the environmental experiences associated with SES. Pubmed, PsycInfo, and Google Scholar databases from January 1, 2000, to December 31, 2019, were systematically searched using terms “Neural” OR “Neuroimaging” OR “Brain” OR “Brain development,” AND “Socioeconomic” OR “SES” OR “Income” OR “Disadvantage” OR “Education,” AND “Early childhood” OR “Early development”. Nineteen studies were included in the full review after applying all exclusion criteria. Studies revealed associations between socioeconomic and neural measures and indicated that, in the first years of life, certain neural functions and structures (e.g., those implicated in language and executive function) may be more sensitive to socioeconomic context than others. Findings broadly support the hypothesis that SES associations with neural structure and function operate on a gradient. Socioeconomic status is reflected in neural architecture and function of very young children, as early as shortly after birth, with its effects possibly growing throughout early childhood as a result of postnatal experiences. Although socioeconomic associations with neural measures were relatively consistent across studies, results from this review are not conclusive enough to supply a neural phenotype of low SES. Further work is necessary to understand causal mechanisms underlying SES-brain associations.

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Socioeconomic Status; Early Childhood; Brain Development

Variability along the socioeconomic status (SES) spectrum is associated broadly with developmental and cognitive indices, academic achievement and educational outcomes, and physical and mental health across the lifespan (Adler & Newman, 2002; Farah, 2017). Although there is no consensus on a single definition of SES, socioeconomic status is a multidimensional construct meant to characterize the degree to which individuals are better or worse off in terms of their access to material and social resources (Adler & Newman, 2002). These resources include access to adequate nutrition, housing, safe and

enriching neighborhoods, income, and education. While the measures that comprise SES may differ cross-culturally, variation in terms of access to resources exists across societies (and species, e.g., Rowell, 1974). Most often, SES measures represent an individual or group's social and economic standing, classified by income and education level (and sometimes occupation) and, more recently, neighborhood qualities, with most of the SES indicators usually being highly (but not perfectly) correlated (Oakes & Rossi, 2003).

Health disparities related to SES have been well-documented in a wide array of physical and mental health conditions in childhood and adulthood, including low birth-weight and preterm birth, heart disease, diabetes, cancer, and schizophrenia (Adler & Newman, 2002; Hakulinen et al., 2020). Further, low-SES is associated with higher rates of mortality, especially in middle adulthood (Adler & Newman, 2002). Children and adolescents from socioeconomically disadvantaged backgrounds are two to three times more likely to develop mental health conditions over the course of their lifetimes (Reiss, 2013), including ADHD (Russell et al.), externalizing behavior problems (Hosokawa et al., 2018), depressive symptoms, substance use problems (Goodman & Huang, 2002), and schizophrenia (Werner, Malaspina, & Rabinowitz, 2007), among others.

Neuroscientific Approach to SES: Given all that is known about how reduced access to social and economic resources can impact one's physical and mental health via diverse sociocultural and neurobiological mechanisms (e.g., access to healthcare, education, exposure to environmental pollutants), some may reasonably question whether it makes sense for neuroscience to approach a problem as fundamentally societal and multifactorial as SES disparities (Farah, 2017). While neuroscience research could not replace sociological and psychological approaches to understanding how SES impacts the lives of individuals, it may offer unique insights into *how* SES-related health disparities manifest and persist (Farah, 2017). Understanding links between various aspects of SES and brain structure and function may increase our understanding of how SES-based health gradients emerge. Specifically, investigating neurobiological correlates of SES-related differences in cognitive function may reveal distinct neural processes or reduced recruitment of the same function underlying these differences. Discernment of these links may contribute to mechanistic understanding of risk and may help inform prevention and intervention programs aimed at ameliorating the effects of socioeconomic disadvantage on health and educational outcomes.

A growing body of research on the neuroscience of SES disparities has revealed consistent associations between SES and cognitive performance, with some neurocognitive systems being especially sensitive to environmental context related to SES (Farah, 2017; Noble et al., 2005). In particular, findings from older children, adolescents, and adults suggest that language, executive function, and declarative memory are among the neurocognitive domains most strongly associated with SES (Fernald et al., 2013; Hart & Risley, 1995; Hoff, 2006; Lawson et al., 2018), with robust evidence relating SES, and particularly maternal education, to children's language skills, reading abilities, and vocabulary. Further, neuroimaging studies have also revealed robust links between SES and brain function. For instance, studies utilizing functional MRI (see Table 1 for glossary of terms) have reported patterns of altered neural activity in school-age children from lower SES backgrounds, including reduced hemispheric specialization for language processing (Raizada et al., 2008) and less efficient functional network organization (Bourne & Rosa, 2006), with similar patterns observed in adults (Chan et al., 2018).

Table 1. Glossary of Terms: Neuroimaging Methods, Neural Measures, and Underlying Neural Constructs.

Method ^a	Description	Brain structure	Main Outcome	Specific Neural Measures and Underlying Neural Constructs ^a
Structural MRI	Magnetic Resonance Imaging (MRI) provides <i>in vivo</i> structural images of the human brain, allowing mapping the <i>anatomy of the brain</i> . MRI measures magnetic signal from hydrogen nuclei in water (H ₂ O) molecules in our brain (or elsewhere in the body); the signal varies in strength depending on the surroundings, providing a means of discriminating between gray matter, white matter and cerebral spinal fluid in structural images of the brain. Variations in the pulse sequences used to acquire MR images can result in image characteristics that emphasize one or more aspects of the tissue or brain anatomy.			Cortical thickness (CT) : combined thickness of the six layers of the cerebral cortex (measured as a local average distance from the white matter surface and the pial surface [i.e., the external cortical surface, which corresponds to pia mater]). Cortical surface area (SA) : the area covered by the cerebral cortex (measured as a surface that runs mid-distance between the white and pial surface, as to not over- or under-represent gyri or sulci dominating in the specific region). Cortical gray matter (regional) volume : the amount of gray matter (including neuronal cell bodies, neuropil [dendrites and unmyelinated axons], and glial cells) that lies between the gray-white interface and the pia mater (measured on MRI images as the volume contained between the white and pial surfaces); cortical volume is, fundamentally, a product of CT and SA. <i>SA and CT are independent, both globally and regionally; each of the three measurements is highly heritable, with SA and CT being under independent genetic influences.</i>
Functional MRI	Functional MRI (fMRI) measures changes in cerebral venous oxygen concentration that <i>correlate</i> with neuronal activity. ^b fMRI takes advantage of the brain's <i>intrinsic</i> signaling, occurring as a result of local increase in concentration of deoxyhemoglobin in venous blood associated with increased neuronal activity (i.e., increased demand for oxygen, which is delivered to neurons by hemoglobin). While measuring local changes in magnetic susceptibility (due to hemoglobin magnetic properties: it is diamagnetic when oxygenated and paramagnetic when deoxygenated, which leads to small differences in the MR signal depending on the degree of oxygenation), fMRI is considered an <i>indirect</i> index of <i>functional brain activation</i> (because the neurovascular coupling mechanism is only partially understood).	Functional physiological change (changes in local concentrations of oxy- and deoxyhemoglobin associated with increased neuronal activity) exploited to map <i>functional brain activity</i>		Changes in BOLD (blood-oxygen-level-dependent) signal , resulting from the change in magnetic field surrounding the red blood cells depending on the oxygen state of the hemoglobin; when BOLD signal changes during a cognitive task are mapped (statistically isolated) in a certain brain region, they are interpreted as an indirect measure of neural activity in that region. <i>Activation, or task-related fMRI is typically used to identify brain regions involved in a specific (motor, cognitive, affective) task.</i>

(Continued)

Table 1. (Continued).

Method ^a	Description	Main Outcome	Specific Neural Measures and Underlying Neural Constructs ^a
Functional connectivity MRI	Derived from conventional fMRI (BOLD) signal, functional connectivity MRI (fcMRI) measures the temporal synchronization of spontaneous BOLD signal fluctuations in different brain regions, putatively reflecting co-activation or <i>functional connectivity</i> between these regions.	<i>Whole-brain functional organization</i> : Temporal correlation between physiological signals, interpreted as functional connectivity	Functional Connectivity (between two brain regions) : statistical dependence among BOLD signal time series recorded in these regions; usually measured with correlations between BOLD signal fluctuations in two brain regions (although alternative methods also exist); founded on the notion that coordinated/synchronous brain activity (across the brain, or among discrete brain regions) reflects a common functional process or functional coupling. <i>Functional connectivity is typically estimated from so-called resting state fMRI acquired in the absence of overt cognitive tasks, during various states of consciousness, including wakefulness and sleep.</i>
Near-Infrared Spectroscopy (NIRS)	Functional Near-Infrared Spectroscopy (NIRS) is an optical imaging technique for detecting local hemoglobin concentration changes. It takes advantage of <i>light scattering changes that accompany cortical activation</i> , especially in the near-infrared part of the electromagnetic spectrum that penetrates the skull much more efficiently than visible light. How much light is reflected from or transmitted through nervous tissue is influenced by changes in optical properties of the illuminated tissue, including local oxygenation. Thus, fNIRS is thought to provide an <i>indirect measure of brain activity</i> .	Functional physiological change (activity-related change in light scattering/tissue reflectance), exploited to <i>map functional brain activity</i>	NIRS signal (intensity of the detected light) : changes in the concentrations of oxygenated (oxy-Hb) and deoxygenated hemoglobin (deoxy-Hb); an indirect measure of neural activity ^b in various brain regions.

(Continued)

Table 1. (Continued).

Method ^a	Description	Main Outcome	Specific Neural Measures and Underlying Neural Constructs ^a
EEG	Electroencephalogram (EEG) is a record of the oscillations of brain electric potentials recorded from sensors on the scalp. The scalp EEG (to be distinguished from intracranial EEG employed in clinical practice for more precise estimation of epileptic locus, for instance) provides robust measures of neocortical <i>dynamic function</i> , detecting <i>changes in electrical activity</i> in the brain <i>on a millisecond-level</i> (comparable to the time frame of neuronal events). The EEG reflects post-synaptic potentials (not action potentials) occurring in cell assemblies of millions of synchronously active neurons, spread through volume conduction between brain and scalp (with the superposition of potentials generated by many local sources). The columnar structure of the cerebral cortex (with large pyramidal cells lined up in parallel) facilitates the electrical summation and passive conduction of the currents produced by large neuronal assemblies.	Monitors brain function by tracking <i>changes in brain electrical activity</i>	Although EEG is a mixture of multiple frequency components, EEG potentials are conventionally described as patterns of activity in five frequency ranges: delta (1–4 Hz), theta (4–7 Hz), alpha (8–12 Hz), beta (13–20 Hz), and gamma (roughly > 20 Hz). Power spectrum (squared amplitude): provides a measure of the energy or variance in the signal as a function of temporal frequency (in each frequency band). <i>Complex mental activity and sustained attention result in increased signal power in the lower frequency ranges (below alpha) and decreased signal power at the higher ranges (alpha and above). Changes in brain state or arousal (sleep, wakefulness, vigilance) also result in modulations of EEG signal.</i>
ERPs	Event-related potentials (ERPs) are scalp EEG recordings that are <i>evoked or event-related</i> , unlike the <i>spontaneous</i> potentials measured with the ongoing EEG. ERPs are the direct responses to some external (or internal, endogenous) stimulation, thought to reflect different stages of information processing. Methodologically, ERPs are extracted from the EEG recordings by time-locking the waveforms with an event, and averaging over multiple events to amplify the ERP from the spontaneous EEG signal. Because of the volume conduction (spreading the currents through brain, skull, and scalp), each ERP component represents a sum of potentials generated from broadly distributed cortical areas.	Monitors brain function by tracking the <i>precise timing</i> (and cortical distribution on scalp) of the <i>neuroelectrical activity generated during cognitive processing</i>	An average ERP waveform consists of a series of waves (voltage fluctuations) defined by peaks and troughs known as ERP components. Amplitude of the ERP component: peak voltage measure within the peak/trough. Latency of the ERP component: timing (in ms) of the component from the stimulus (or event) onset. <i>Measurement of changes in the amplitude and timing of peaks of the ERP components (known to be sensitive to certain cognitive functions or task conditions) allows inferences about the sequence and timing of information processing (e.g., stimulus evaluation and attention allocation [p3], response evaluation [ERN], etc.).</i>

^aOnly methods and measures used in the studies included in the synthesis are described. This is not an exhaustive list of brain imaging techniques or possible neural measures.
^bIn functional imaging methods (fMRI, fNIRS), the term *increased brain activity* or *increased neuronal activity* cannot be equated with increased neuronal firing rate or any other electrophysiologically defined terms. Rather, in fMRI or fNIRS, which measure vascular or light scattering changes in response to neuronal activity, any change in brain activity inducing an increase in local blood flow/light reflectance is taken to indicate an *activation*.

Beyond language, parental SES also appears to mediate the links between brain activity in regions supporting spatial processing and gains in math skills (Demir-Lira et al., 2016).

In addition to functional brain correlates of SES, there is growing evidence showing links between SES and brain structure, including white matter organization and brain morphology indexed with regional volumes, cortical thickness, or surface area (see Table 1 for glossary of terms). Indeed, SES has been found to correlate with many aspects of brain structure and cortical maturation in school-age children, adolescents, and adults, including regional volumes, surface area, and cortical thickness (Gianaros et al., 2008, 2015; Mackey et al., 2015; Noble et al., 2015). For example, Noble et al. (2015) reported a logarithmic relationship between household income and brain surface area in children between ages 3 and 20 years. Namely, for children from lower-income households, small incremental increases in income were associated with larger increases in brain surface area, compared to children from higher-income families, for whom similar income increments were associated with smaller differences in surface area, suggesting that income may relate more strongly to brain structure in the most disadvantaged children. These effects were particularly evident in brain regions implicated in language, executive function, and spatial skills (Noble et al., 2015).

SES and Early Childhood: Despite the growing literature on neural correlates of socioeconomic disparities across the lifespan, less is known about the links between SES and brain development in *early* childhood, during the critical window when the brain undergoes profound maturational changes (Tau & Peterson, 2010) and exhibits peak plasticity (Kolb & Gibb, 2011). Corollary to the maximal neuroplasticity, the first years of life are a period of rapid learning and remarkable advances in cognitive and behavioral development (Bornstein, 2013). The time from birth to five years is when many foundational skills across all areas of development are acquired, including gross and fine motor skills, receptive and expressive language skills allowing communication with others, and socioemotional skills permitting the child to learn from others (caregivers and peers alike) through social learning and to form relationships with others (Bornstein, 2013). These skills serve as developmental building blocks, laying foundation for acquisition of academic and socioemotional competencies in middle and late childhood, with lifelong effects on one's function, health, and well-being (Shonkoff et al., 2012). At the same time, both brain and neurocognitive development in early childhood are particularly responsive to environmental input (Merz et al., 2019; Tierney & Nelson, 2009), setting the stage for the greatest window of opportunity for modifying developmental trajectories (cf. Dawson et al., 2012).

Our understanding of brain development in early childhood (from birth to age 5 years) has expanded greatly over the past several decades (Gilmore et al., 2018; Stiles & Jernigan, 2010). While certain brain regions, like sensory and motor cortices, develop earlier and more rapidly than others (Tau & Peterson, 2010), cortical and subcortical gray matter volumes, in general, undergo the most robust growth in the first year of life, with additional growth in the second year, but minimal increases after age 2 (Gilmore et al., 2012). This rapid volume growth likely reflects continuous synapse formation in the first 2 years of life, albeit with variable rates across different brain regions (Huttenlocher & Dabholkar, 1997). Beyond the volumetric growth, maturational changes in brain structure in the first years of life are marked by increasing cortical thickness as well as expansion of cortical surface area (Brown & Jernigan, 2012; Lyall et al., 2015). While

most white matter tracts are already in place at birth, their fiber connections undergo rapid myelination during infancy and toddler years, becoming more functionally efficient in information transfer. Myelination of white matter tracts peaks in the first year of life, with less rapid changes during toddlerhood, and slower changes thereafter, well into adulthood, although there are considerable regional variations in myelination rate across different white matter bundles (Dubois et al., 2014; Paterson et al., 2006).

Concurrent with the rapid macro-scale cortical and white matter maturation (generally peaking in the first 2 years of life, regional differences notwithstanding) is an equally rapid development of brain functional brain networks. Most functional networks are present at birth and undergo reorganization and fine-tuning throughout early childhood (Gao et al., 2015), in parallel with the behavioral and cognitive milestones achieved during this pivotal period in human development (Johnson, 2001). In particular, primary sensory and motor networks – implicated in processing sensory information and supporting motor development – become increasingly more integrated in the first year of life and substantially resemble adult topology by age two (Gao et al., 2015; Lin et al., 2008), while supra-modal functional networks implicated in higher-order cognitive functions are far from the adult-like organization in the first postnatal years and undergo prolonged maturation over the first decades of life (Dosenbach et al., 2010; Fair et al., 2008; Gao et al., 2015; Hoff, Van den Heuvel, Benders, Kersbergen, & De Vries, 2013).

Critically, these neurodevelopmental processes supporting remarkable neurobehavioral maturation unfold during the time in a child's life when family and home context are the most influential environment affecting development, in contrast to middle and late childhood and adolescence when school and peers become primary environmental influences on development (Duncan et al., 2012). While it is recognized that the circumstances associated with low family SES in early life can have broad-reaching, long-term impacts on developmental outcomes such as linguistic skills and literacy (Reardon et al., 2013), relatively few studies have focused on the links between SES and neurodevelopment in early childhood, in large part because of the practical difficulties associated with acquiring brain imaging data in young children. Additionally, individuals from lower-resourced communities have lower rates of research participation given the socioeconomic barriers they are facing, such as inadequate access to transportation, limited opportunities for time off work, etc. Despite these challenges, a growing body of research has begun highlighting associations between SES and neurodevelopment in young children (ages 0–5 years), with some recent reports suggesting that the magnitude of the effect of maternal education on children's neurocognitive outcomes at preschool age is as large as that of having experienced a brain injury at birth (Benavente-Fernández et al., 2019; Hung et al., 2015).

Given the importance of the early years for laying the foundation for developing cognitive, social, and emotional skills, and the remarkable maturational changes that the brain undergoes during this pivotal developmental period (Gilmore et al., 2018), the present review aims to examine the available evidence for associations between SES and brain development in early childhood (ages 0–5 years), addressing the following questions: Are there neural correlates of SES in early childhood? If so, are they general or specific to certain brain regions or functions? Do they vary on an SES gradient, or is there a threshold at which point SES effects can be observed (e.g., poverty)?

Methods Search Strategy and Identification of Studies

This systematic review was conducted in alignment with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher, 2009), with the PRISMA flow diagram outlining the process provided in Figure 1. Three electronic databases or search engines (PubMed, PsychINFO, and Google Scholar) were searched systematically between December 2019 and January 2020 to identify studies investigating associations between SES measures and brain development in early childhood (0–5 years of age). Search keywords included “Neural” OR “Neuroimaging” OR “Brain” OR “Brain development,” AND “Socioeconomic” OR “SES” OR “Income,” AND “Early childhood” OR “Early development”. Search results containing keywords in the title or abstract were further restricted to peer-reviewed articles written in the English language, with no limiters applied with regard to date of publication. Finally, reference sections from eligible articles retained after abstract review were consulted for articles not included in either of the database searches. A subsequent search (referred henceforth as the second search) was conducted in September 2020, including all the aforementioned search keywords, as well as terms “Disadvantage” OR “Education,” following a suggestion by

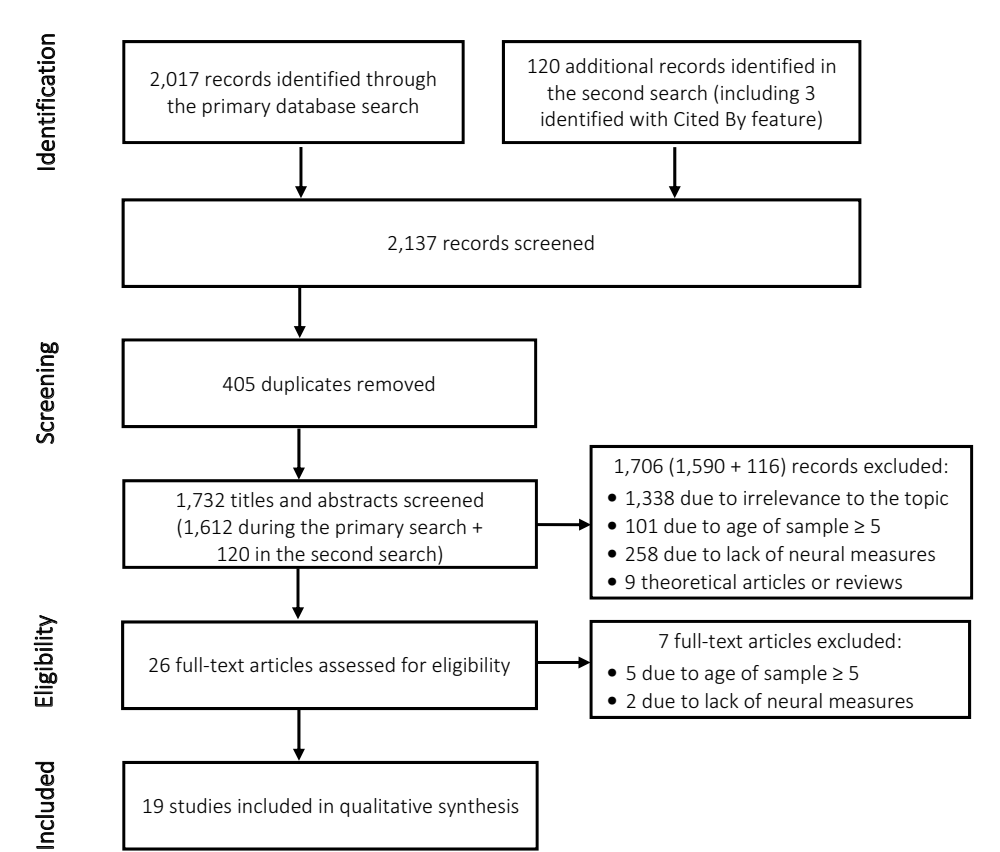


Figure 1. PRISMA Flow chart outlining identification, screening, eligibility, and inclusion of records.

an anonymous reviewer. Similar to the first search, results containing search keywords in the title or abstract only were returned. This second search also included the “Cited by” feature in Google Scholar for each of the articles retained for review.

Article Screening and Selection

After removal of duplicates, articles’ titles were first screened by authors L.O. and B.C. (separately) for relevance to early childhood development, SES, and neural measures (measurements of brain structure or function). Disagreements were resolved via discussion and consensus was reached in each case. Next, remaining article abstracts were screened for eligibility by L.O. and B.C. based on meeting each of the following three inclusion criteria: sample of children aged 5 years or younger (with or without psychiatric or neurodevelopmental disorders), measurement of SES (including a measure of income, education, or neighborhood characteristics, or some combination of these), and measurement of brain structure or function (using anatomical, functional, or diffusion-weighted MRI, functional near infrared spectroscopy [fNIRS], or EEG/ERPs; see [Table 1](#) for glossary of terms). The following exclusion criteria were also implemented: the article being a review or a theoretical article, and the study sample including children with a broad age range encompassing but not limited to early childhood (participants younger and older than 5 years). Across both primary and second searches, the title screening, followed by the abstract screening resulted in 26 articles retained for the full-text eligibility review, with 7/26 articles excluded from the qualitative synthesis at this stage (see [Figure 1](#) for exclusion reasons), and 19 articles retained for the full review and rated for study quality.

Study Quality Assessment

The articles included in the synthesis were rated for quality by authors L.O. and B.C. on a 0-to 6-scale designed to assess the overall methodological quality and potential risk of bias, with one point assigned for each of the following criteria: 1) reporting of demographic information of sample, and including sample representativeness (i.e., not all participants of one racial group or gender), 2) study design: longitudinal studies were given one point while cross-sectional studies received 0 for this criterion, 3) sample size: sample included at least 20 participants in each socioeconomic strata (if stratified), or at least 20 individuals who could be characterized by the authors as lower SES, and 20 who could meet criteria defined by study authors for mid- to high-SES, 4) neuroimaging studies testing a priori (vs. post-hoc/exploratory) hypotheses, 5) reporting and controlling for (or matching on) relevant covariates (gender, age, ethnicity, exposure to multiple language, exposure to substances in utero, prematurity), and 6) including more than one measure of SES (e.g., parental education AND income). Thus, a score of 0 on the Quality Rating signifies that the article did not meet any of these basic criteria, indicating relatively low quality and higher risk of bias, whereas a score of 6 corresponds to fulfilling all criteria, indicating relatively high quality and lower risk of bias.

Results

The primary search conducted between December 2019 and January 2020 yielded 2,017 studies. After removing 405 duplicates, the remaining 1,612 studies were screened at the title and abstract level. Title and abstract screening resulted in the exclusion of 1,590 articles for irrelevance to the topic of the present review (e.g., early development of the *heart*, socioeconomic health gradients in adulthood, socioeconomic disadvantage and *behavioral* problems in early childhood, etc.; $n = 1,297$), inclusion of children outside of the age range of 0–5 years ($n = 84$), lack of neuroimaging measures ($n = 205$), or because they were review articles on related topics ($n = 4$). The second search conducted in September 2020 (including all the aforementioned search keywords, as well as “Education” OR “Disadvantage”) yielded 120 additional articles (after removing duplicates of the articles included in the previous search), including 3 articles identified through Google Scholar “Cited By” feature. Title and abstract screening of these 120 articles resulted in the exclusion of 116 for irrelevance to the topic of the present review (e.g., socioeconomic correlations with learning or achievement; $n = 41$), inclusion of children outside of the age range of 0–5 years ($n = 17$), lack of neuroimaging measures ($n = 53$), or because they were review articles on related topics ($n = 5$).

The resultant twenty-six full-text articles were reviewed, with seven excluded for inclusion of children older than 5 years of age ($n = 5$) or not including neural measures ($n = 2$). Ultimately, 19 studies were selected for the full systematic review (see [Figure 1](#)).

A summary of the 19 reviewed articles is provided in [Table 2](#), including neuroimaging techniques used, SES variables measured, and summarized results. A summary of ratings for quality criteria is provided in [Table 3](#). Each of the 19 studies included children under the age of 5 from a broad range of socioeconomic backgrounds and utilized measures of brain structure or function (i.e., anatomical or functional MRI, functional near infrared spectroscopy [fNIRS], or EEG/ERPs; see [Table 1](#) for glossary of terms). Of the 19 studies, five used longitudinal designs. Each study characterized SES differently, depending on regional and cultural variability in social and economic patterns. As such, “high” and “low” SES refer to *relative* socioeconomic circumstances and may not generalize across samples. The summary of the 19 studies reviewed below is organized by the investigated effects on brain structure and function, and by approximate chronological age of study participants within each section.

SES and Brain Structure in Early Childhood

Of the 19 studies reviewed, six focused on structural brain correlates of SES in early childhood (i.e., brain volume, cortical thickness, surface area). One study examining the impact of SES on brain development in a birth cohort of 44 (25 low-SES, 19 high-SES) African American female neonates reported correlations between SES and gray matter volume as early as 5 weeks postnatally, with lower SES indices associated with smaller cortical (cortex of both hemispheres and hippocampi) and deep (thalami and basal ganglia) gray matter volumes (Betancourt et al., 2016). Income-to-needs ratio (a ratio of household income to the federal poverty level, with an income-to-needs ratio of 1

Table 2. Summary of Results and Quality Ratings.

Title	Authors	Design	Sample Size (sex: F, M)	Age (range or mean)	SES Assessment	Continuous vs. Discrete	Neural Measures	Summary Results	Quality Rating (0–6)
Effect of socioeconomic status (SES) disparity on neural development in female African-American infants at age 1 month	Betancourt, L., Avants, B., Farah, M., Brodsky, N., Wu, J., Ashtari, M., Hurt, H.,	Birth cohort of full-term African-American female infants	44 (all F)	Neonates: 4–6 weeks	Maternal education and income-to-needs ratio	Dichotomous (high vs. low SES)	Structural MRI	Lower SES was associated with smaller cortical and deep gray matter volumes.	3
	Associations among family socioeconomic status, EEG power at birth, and cognitive skills during infancy	Longitudinal (sample drawn from a larger longitudinal study on prenatal exposures and birth outcomes)	66 (28 F, 29 M)	Neonates (12–96 hours after birth): mean gestational age: 38 weeks	Parental education, income-to-needs ratio	Continuous	EEG	No relationship between neonatal EEG measures and SES variables (nor between SES and memory or language outcome variables at 15 months of age).	
Maternal behavior and socioeconomic status predict longitudinal changes in error-related negativity in preschoolers	Brito, N., Fifer, W., Myers, M., Elliott, A., Noble, K.	Longitudinal	119 (69 F, 50 M)	3.5 years (first timepoint); 4.5 years (second timepoint)	Four-factor Hollingshead Index	Continuous	EEG/ERP	No association between ERN amplitude and SES, cross-sectionally; however, the developmental change in the ERN (between ages 3 and 4) was observed only in children from higher SES households and whose maternal sensitivity was rated as “high.”	5

(Continued)

Table 2. (Continued).

Title	Authors	Design	Sample Size (sex: F, M)	Age (range or mean)	SES Assessment	Continuous vs. Discrete	Neural Measures	Summary Results	Quality Rating (0–6)
Frontal theta activation associated with error detection in toddlers: influence of familial socioeconomic status	Conejero, A., Guerra, S., Abundis- Gutiérrez, A., Rosario Rueda, M.	Cross-sectional	52 (26 F, 26 M)	16–18 months	Parental education, income-to-needs ratio, occupation	Continuous	EEG/ERP	Low-SES toddlers showed reduced error-related negativity (ERN), an ERP index of self-monitoring and executive control, compared to higher-SES toddlers.	4
Functional network development during the first year: Relative sequence and socioeconomic correlations	Gao, W., Alcauter, S., Elton, A., Hernandez- Castillo, C., Smith, J., Ramirez, J., Lin W.	Longitudinal	65 (35 F, 30 M)	1–12 months	Four-factor Hollingshead Index	Continuous	Functional MRI/ functional connectivity	Moderate positive correlations between SES variables (income and maternal education) and within- network functional connectivity of the default mode (DMN) and sensorimotor networks (an index of network maturation) at age 6 months.	5
Family poverty affects the rate of human infant brain growth	Hanson, J., Hair, N., Shen, D., Shi, F., Gilmore, J., Wolfe, B., Pollak, S.	Longitudinal	77 (31 F, 46 M)	1 month to 4.5 years	Maternal education and household income	Categorical (low, middle, high); based on income proportional to federal poverty level	Structural MRI	Low SES children had significantly lower total gray matter volumes and lower average frontal and parietal gray matter volumes. Slowed gray matter growth trajectories for children from low SES households were also reported.	6

(Continued)

Table 2. (Continued).

Title	Authors	Design	Sample Size (sex: F, M)	Age (range or mean)	SES Assessment	Continuous vs. Discrete	Neural Measures	Summary Results	Quality Rating (0–6)
Impact of demographic and obstetric factors on infant brain volumes: A population neuroscience study	Knickmeyer, R., Xia, K., Lu, Z., Ahn, M., Jha, S., Zou, F., Zhu, H., Styner, M., Gilmore, J.	Cross-sectional	756 (350 F, 406 M)	8–13 months (mean: 9.7 months)	Parental education, income	Continuous	Structural MRI	Parental education had marginal (positive) associations with infant brain volumes, whereas income did not. The effects of parental education were partially mediated by differences in birthweight.	4
Associations between cortical thickness and reasoning differ by socioeconomic status in development	Leonard, J., Romeo, R., Park, A., Takada, M., Robinson, S., Grotzinger, H., Last, B., Finn, A., Gabrieli, J., Mackey, A.	Cross-sectional	115 (gender ratio not reported)	4–7 years	Maternal education, income	Continuous and dichotomous measures	Structural MRI	In young children from lower SES backgrounds, greater cortical thickness of the RLPGC was related to better reasoning ability. There was an age-related increase in RLPGC cortical thickness (indicating protracted, slower development of this region) in lower SES children with higher reasoning ability, but not in lower SES children with lower reasoning ability, or in higher SES children.	5
Socioeconomic disparity in prefrontal development during early childhood	Moriguchi, Y., & Shinohara, I.	Cross-sectional	93 (48 F, 45 M)	Mean: 4.9 years	Maternal education and household income	Continuous and dichotomous measures	Functional near infrared spectroscopy (fNIRS)	Continuous measures of SES were not correlated with fNIRS activation on an executive function task. However, in the dichotomous SES analyses (poverty vs. no poverty), children in the no-poverty group exhibited significant fNIRS activation, whereas children who were experiencing poverty did not.	4

(Continued)

Table 2. (Continued).

Title	Authors	Design	Sample Size (sex: F, M)	Age (range or mean)	SES Assessment	Continuous vs. Discrete	Neural Measures	Summary Results	Quality Rating (0–6)
Poverty, cultural disadvantage and brain development: A study of pre- school children in Mexico	Otero, G. A.	Cross-sectional	42 (20 M, 22 F)	4 years (range not reported)	Sociocultural questionnaire: Score cutoff based on parental education, occupation, number of inhabitants in child's home income, and nutrition	Dichotomous (high vs. low SES)	EEG	The low SES group (referred to as "high risk children") had greater delta absolute power in the frontal and left central leads, as well as greater total absolute power in frontal, central, and parietal leads, than the higher SES children.	3
Effects of antenatal maternal depressive symptoms and socio-economic status on neonatal brain development are modulated by genetic risk	Qui, A., Shen, M., Buss, C., Yap- Seng, C., Kwek, K., Saw, S., Gluckman, P., Wadhwa, P., Entringer, S., Styner, M., Karnana, N., Heim, C., O'Donnell, K., Holbrook, J., Fortier, M., Meaney, M.	Cross-sectional	253 (115 F, 138 M)	Neonates: 0–1 months (mean gestational age: 38 weeks)	Household income	Continuous	Structural MRI	Genomic profile risk score for major depressive disorder (GPRS-MDD) moderated the negative association between SES and right amygdala and hippocampal volumes and shapes.	4
Socioeconomic status predicts hemispheric specialization of the left inferior frontal gyrus in young children	Raizada, R., Richards, T., Meltzoff, A., Kuhl, P.	Cross-sectional	14 (7 F, 7 M)	Mean: 5.3 years (all preschool children)	Four-factor Hollingshead Index	Continuous	Functional MRI	Interhemispheric difference in activation of the inferior frontal gyrus (in response to rhyming task) was strongly correlated with SES.	2

(Continued)

Table 2. (Continued).

Title	Authors	Design	Sample Size (sex: F, M)	Age (range or mean)	SES Assessment	Continuous vs. Discrete	Neural Measures	Summary Results	Quality Rating (0–6)
Brain connectivity and socioeconomic status at birth and externalizing symptoms at age 2 years	Ramphal, B., Whalen, D., Kenley, J., Yu, Q., Smyser, C., Rogers, C., Sylvester, C.	Longitudinal (but cross- sectional imaging data)	112 (66 F, 46 M); full- term n = 75; pre-term n = 37	Neonates: Mean postmenstrual age for full-term infants: 39.3 weeks; mean postmenstrual age for pre-term infants: 37.5 weeks	Insurance type (public vs. private), area deprivation index, and a “composite score”	Continuous and dichotomous measures	Functional MRI	Lower SES was associated with altered (increased) fronto- striatal connectivity.	6
Beyond the 30- Million-Word Gap: Children’s conversational exposure is associated with language- related brain function	Romeo, R., Leonard, J., Robinson, S., West, M., Mackey, A., Rowe, M., Gabriele, J.	Cross-sectional	36 (14 F, 22 M)	Mean: 5.8 years (4–6 years)	Parental education	Continuous	Functional MRI	Caregiver-child conversational turns and Broca’s area activation both mediated the relationship between parental education and children’s language scores.	3
Income, neural executive processes, and preschool children’s executive control	Ruberry, E., Lengua, L., Crocker, L., Bruce, J., Upshaw, M., Somerville, J.	Cross-sectional	119 (gender ratio not reported)	4.5–5.6 years	Income	Categorical (poverty, low income, and mid- to high- income)	EEG/ERP	Income was unrelated to ERP measures (N2 and P3) of executive control, although both income and ERP measures were related to performance on an executive function task performed by preschoolers.	4

(Continued)

Table 2. (Continued).

Title	Authors	Design	Sample Size (sex: F, M)	Age (range or mean)	SES Assessment	Continuous vs. Discrete	Neural Measures	Summary Results	Quality Rating (0–6)
Prenatal socioeconomic status and social support are associated with neonatal brain morphology, toddler language and psychiatric symptoms	Spann, M., Bansal, R., Hao, X., Rosen, T., Peterson, B.	Longitudinal (but cross- sectional imaging data)	37 (13 F, 24 M)	Neonates at time of MRI scan (1–6 weeks postnatally), toddlers (24 months) at behavioral assessment	Four-factor Hollingshead Index	Continuous	Structural MRI	Low SES was associated with greater local volumes in occipital, temporal, and frontal regions. However, having a partner (an index of social support) moderated these effects (partnered status was associated with lower volumes than non-partnered status among low-SES infants).	5
Socioeconomic status and neural processing of a go/no-go task in preschoolers: An assessment of the P3b	St. John, A., Finch, K., Tarullo, A.	Cross-sectional	105 (52 F, 53 M)	4.5–5 years	Income-to-needs ratio and parental educational level	Continuous	EEG/ERP	Higher household income was related to larger P3 amplitudes (an ERP index of attention allocation and executive control).	5
Socioeconomic status and functional brain development – associations in early infancy	Tomalski, P., Moore, D., Ribeiro, H., Axelsson, E., Murphy, E., Karmiloff-Smith, A., Johnson, M., Kushnerenko, E.	Cross-sectional	41 (31 F, 14 M)	Mean: 7.5 months	Household income, maternal and paternal education and occupation	Dichotomous (high vs. low SES)	EEG	Low SES infants showed lower frontal gamma power than higher-SES infants.	5

(Continued)

Table 2. (Continued).

Title	Authors	Design	Sample Size (sex: F, M)	Age (range or mean)	SES Assessment	Continuous vs. Discrete	Neural Measures	Summary Results	Quality Rating (0–6)
The relationship between biological and psychosocial risk factors and resting-state functional connectivity in 2-month-old Bangladeshi infants: A feasibility and pilot study	Turesky, T., Jensen, S., Yu, X., Swapna, K., [...], Nelson, C., Gaab, N.	Cross-sectional	32 (15 F, 17 M)	2–3 months	Income-to-needs ratio, maternal educational level	Dichotomous (below global poverty level vs. above poverty level)	fMRI	Infants from extreme poverty had altered (less negative) functional connectivity between amygdala and precuneus than infants with relatively greater access to resources.	4

Table 3. Article Characteristics for Quality Ratings.

Article	Representative sample	Longitudinal design	Sample size (n ≥ 20)	A priori hypothesis testing	Inclusion of covariates	Multiple SES measures	Total rating
Betancourt et al., 2016				+	+	+	3
Brito et al.,		+	+	+	+	+	5
Brooker, 2018	+	+	+	+		+	5
Conejero et al., 2018			+	+	+	+	4
Gao et al., 2015		+	+	+	+	+	5
Hanson et al., 2013	+	+	+	+	+	+	6
Knickmeyer et al., 2017	+		+	+	+	+	4
Leonard et al., 2019	+		+	+	+	+	5
Moriguchi & Shinohara, 2019	+		+	+		+	4
Otero, 1997			+	+		+	3
Qiu et al., 2017	+		+	+	+		4
Raizada et al., 2008				+		+	2
Ramphal et al., 2020	+	+	+	+	+	+	6
Romeo et al., 2018			+	+	+	+	3
Ruberry et al., 2016	+		+	+	+		4
Spann et al., 2020	+	+		+	+	+	5
St. John et al., 2019	+		+	+	+	+	5
Tomalski et al., 2013	+		+	+	+	+	5
Turesky et al., 2019	+			+	+	+	4

signifying income at the federal poverty threshold), and maternal income were used to index SES. While the inclusion of only African American children removed race/ethnicity as a possible confound of SES, it has limited the generalizability of these findings.

Further evidence of links between SES and brain volumes in infancy comes from a study conducted on 756 8–12-month-old infants from varying socioeconomic backgrounds (Knickmeyer et al., 2017) reporting positive associations between parental education and total white matter and gray matter volumes. However, the effects of parental education on brain volumes were mediated in part by birthweight. Because no effect of household income on brain volumes was observed, the authors reasoned that education associations with brain volumes could not be explained by differences in financial resources, but may rather reflect differences in “cultural capital,” with lower education representing reduced awareness of the importance of prenatal care, proper nutrition, and avoidance of environmental pollutants, all of which could impact prenatal brain development and birthweight (Knickmeyer et al., 2017).

Results from a study investigating links between brain structure and SES in a cohort of 77 infants followed longitudinally from ~1 month to 4.5 years of age (Hanson et al., 2013) align well with these findings. Namely, the authors reported that low SES children have lower total gray matter volumes than higher-SES children, and that the SES effect was also observed in the gray matter growth trajectories measured longitudinally in a subset of the participants. The authors concluded that SES influences the rate of human infant brain growth, and these differences were not accounted for by infant birthweight, early health, or head size at birth. Consistent with the findings by Betancourt et al. (2016), Hanson et al. (2013) found no SES-associations with total cerebral volume or white matter volume. However, in contrast to Betancourt et al. (2016), the authors used a dichotomous SES variable (i.e., high vs. low SES), indexed by an income to needs ratio at or below poverty (low SES) vs. above the federal poverty line, with a parent who had at least a high-school education (higher SES).

In contrast to the positive associations between SES and total brain volumes or volumes of broad tissue-based brain segments (cortical gray, deep gray, or total white matter) reported in the studies reviewed heretofore, investigations focusing on targeted brain regions appear to reveal negative links between regional brain morphology and SES. Qiu et al. (2017) investigated the relationship between household income, prenatal maternal depressive symptoms, and cortical volumes and thickness in targeted brain regions implicated in depression (amygdala, hippocampus, and orbitofrontal and ventromedial prefrontal cortex) in a birth cohort of neonates born at term and scanned shortly after birth (mean post-conception age at scan 40 weeks). They reported negative associations between household income, measured at 26 weeks of gestation, and right amygdala and hippocampal volumes (measured postnatally), although this relationship was moderated by genomic profile risk scores for major depressive disorder (GPRS-MDD; Qiu et al., 2017). That is, neonates with genetic profiles associated with heightened risk for developing depression showed a negative relationship between their family income and right amygdala and hippocampal volumes, whereas those with low genetic risk profiles showed no such association. The authors interpreted these findings as evidence of gene–environment interdependence in the fetal and peri-natal development of brain regions implicated in emotional function.

Similarly, examining SES-based differences in brain morphology in healthy neonates ($n = 37$) scanned between the 1st and 6th weeks of postnatal life, using the four-factor Hollingshead index (Hollingshead, 1975), a composite measure consisting of marital togetherness vs. separation, occupational status, education, and income to index SES, Spann and colleagues found that infants born to mothers with lower SES had larger (surface-based) local volumes in the right occipital lobe, left temporal pole, and left inferior frontal and anterior cingulate regions (Spann et al., 2020). However, partner status (single vs. partnered) moderated these effects in some regions, such that partnered status was associated with lower volumes than non-partnered status among low-SES infants. According to the authors, these findings suggest that social support (measured via partner status) may buffer the effects of early life low SES on infant brain structure.

Because the relationship between cortical thinning and cognitive and developmental skills within lower-SES children is not well understood, Leonard et al. (2019) investigated whether the relationship between reasoning abilities (indexed with the matrix reasoning subscale of the Wechsler Preschool and Primary Scale of Intelligence) and cortical thickness varied by SES in early childhood (ages 4–7 years; $n = 115$) and in adolescence (ages 12–16 years; $n = 59$). Reviewing the early childhood results only, cortical thickness of the rostro-lateral prefrontal cortex (RLPFC), part of the distributed frontoparietal network, was differentially related to reasoning abilities by SES. Specifically, thicker RLPFC cortex was related to better reasoning abilities in young children from lower SES backgrounds (with SES indexed by maternal education level), but not in children from higher SES families. Further, only children from lower SES backgrounds with strong reasoning skills showed a positive relationship between RLPFC thickness and age, while those from lower SES families with lower reasoning skills showed an age-associated (but non-significant) decline or thinning of the RLPFC. No such interaction between age, reasoning ability, and cortical thickness was found in the higher-SES group. The authors interpreted these results as reflecting possible compensatory (or adaptive) mechanisms in response to life circumstances associated with lower SES, such that for

children from lower-SES environments slower neural development during early childhood (protracted RLPFC thickening instead of normative thinning at this age) – allowing the brain to be more flexible and responsive to learning – may promote the development of strong reasoning skills (Leonard et al., 2019).

SES and Brain Function in Early Childhood

In addition to examining the links between SES and brain structure, a growing number of studies have focused on brain functional correlates of SES in early development, using functional [connectivity] MRI (f[c]MRI), functional near infrared spectroscopy (fNIRS), and EEG, all used to assess neural function with brain activity or connectivity metrics (see Table 1 for glossary of terms on these methods and measures).

In a study investigating links between SES and early postnatal brain connectivity (Ramphal et al., 2020), resting state fMRI data were acquired during natural sleep in neonates born at term ($n = 75$) and term-equivalent newborns born prematurely (gestational age at birth < 30 weeks; $n = 37$), with both groups scanned at an equivalent post-menstrual age of ~ 38 –39 weeks. The authors used three indices to characterize SES in the sample: insurance type (public vs. private), a neighborhood deprivation index, and an aggregate variable based on insurance type, maternal race, maternal age, single-parent status, and maternal education. Functional connectivity in circuits implicated in psychiatric illness was targeted, including connectivity between striatum, medial prefrontal cortex (mPFC), ventrolateral prefrontal cortex (vlPFC), and dorsal anterior cingulate cortex (dACC). The results revealed that, in neonates, socioeconomic disadvantage (indexed with all three SES measures) was associated with increased fronto-striatal connectivity (between striatum and mPFC) and with decreased local vlPFC connectivity, controlling for preterm birth (Ramphal et al., 2020). These findings suggest that the impact of socioeconomic disadvantage on development of the brain systems implicated in externalizing symptoms (fronto-striatal circuitry) is already evident at birth.

In an investigation focusing on low-resource setting conducted in Bangladesh, Turesky and colleagues (2019) acquired functional MRI in naturally asleep infants (age ~ 2.5 months) aiming to estimate neonatal functional connectivity (FC). The authors reported that infants from “extremely poor families” ($n = 16$) showed greater (i.e., less negative) FC between bilateral amygdala and precuneus, compared to infants from relatively more affluent families ($n = 16$; Turesky et al., 2019). Given that *attenuation* in typically negatively-correlated amygdala and precuneus is associated with mood disturbances in children (Barch et al., 2016), this finding suggests that severe economic hardship may be a risk factor for altered amygdala function and connectivity (and future psychopathology), observed as early as the first months of life. Notably, amygdala/precuneus FC was also associated with prenatal family conflict, above and beyond the poverty level, suggesting that this psychosocial risk factor commonly associated with poverty may also be linked to altered amygdala circuitry. Importantly, these results represent the first MRI study of its kind carried out in a low-resource setting with families experiencing “extreme poverty” (defined as living below the global poverty line), demonstrating feasibility for conducting such studies in the future. Overall, in line with the pattern observed by Ramphal et al. (2020), these results indicate that the

relationship between poverty and brain functional connectivity can be observed in the first months of life.

Aimed at examining maturational trajectories of functional brain networks during the first year of life, Gao et al. (2015) obtained longitudinal functional MRI data, acquired during natural sleep in infants born at term and scanned every 3 months, from birth to 12 months of age. The authors observed moderate correlations between SES (measured with income and maternal education) and within-network functional connectivity in the default mode network (DMN) and sensorimotor networks in infants 6 months of age, with higher income and maternal education associated with greater within-network connectivity in these networks. Additionally, higher income was associated with lower outside-network connectivity in the DMN at 6 months of age. There were no significant effects detected at any other age points (between 0 and 12 months of age), nor significant effects of SES on the longitudinal growth rates of any network. The authors interpreted these findings as evidence of the association between higher SES and greater network maturation (evinced by stronger within-network connectivity and weaker connectivity between DMN and other networks) in the first year of life, and particularly at 6 months of age, pointing to the unique importance of this age in the expression of SES effects on functional connectivity (Gao et al., 2015).

SES correlations with brain function have also been observed in studies using EEG or Event-Related Potentials (ERPs). In an investigation of neonates (studied at 12–96 hours after birth), no associations between socioeconomic variables (parental education and income-to-needs ratio) and cortical activity, indexed with resting EEG power, were found, for any of the frequency bands or any scalp locations (Brito et al.,). However, in another study, differences between infants from low- and high-income families were reported at age 6–9 months, with lower frontal gamma power observed in infants from low-income homes (Tomalski et al., 2013). Although the functional meaning of high-frequency resting brain oscillations (gamma waves) is still debated, gamma-band activity in frontal regions has been associated with language skills in toddlers and working memory in adults, and thus the effects of socioeconomic disparities on gamma activity in early infancy may indicate early risk for language and attention difficulties (Tomalski et al., 2013).

Further evidence of SES-related patterns of brain EEG activity comes from a study of 56 toddlers ages 16 to 18 months (Conejero et al., 2018). The authors investigated the developing neural mechanisms supporting executive control and self-regulation using the error-related negativity (ERN) ERP component. The ERN can be observed in adults and older children after error commission, or in response to perceived errors. In infants and toddlers, the ERN is thought to reflect the emergence of self-regulation or self-monitoring, and has been observed in infants as young as 9 months of age in response to simple arithmetic errors committed by puppets (Berger et al., 2006). The results revealed substantial differences in the brain response to errors in toddlers from higher- vs. lower-SES backgrounds. In particular, lower-SES toddlers (defined here according to parental education level) showed reduced ERN responses to observing errors (e.g., in a simple age-appropriate puzzle completion, observing a chicken body being completed with an elephant head), corresponding to less mature neural index of self-monitoring and executive attention, compared to higher-SES peers.

In line with these findings, Brooker (2018) also reported SES-ERN relationships in preschool children. In a longitudinal design, aiming to examine ERN development between ages 3 and 4 years, SES was assessed using the Hollingshead index, and a go/no-go task was used to elicit ERN. A measure of parenting style or parental sensitivity (hypothesized to be a proximal factor related to ERN development, in contrast to SES, thought to be a distal environmental factor linked to developmental changes in ERN) was also obtained, using the Coping With Children's Negative Emotions Questionnaire. The results revealed no cross-sectional associations between ERN amplitude and SES in 3- or 4-year-old children (and no significant change in ERN amplitude between ages 3 and 4). However, a 3-way interaction between ERN, SES, and maternal sensitivity ratings was detected, such that development of the ERN (the expected increase in amplitude between ages 3 and 4) was observed only for children from high SES backgrounds and whose mothers reported high sensitivity (Brooker, 2018). These results are consistent with other findings reviewed here suggesting that low SES may compromise early brain development (e.g., Hanson et al., 2013) and the development of the neural index of executive control (ERN) in particular (Conejero et al., 2018).

Yet another investigation of socioeconomic relations with ERP measures used a go/no-go task to examine the P3 ERP component, a putative index of executive function involving inhibition and attention allocation (St. John et al., 2019). The authors reported that higher income-to-needs ratio (but not parental education) was associated with larger P3 amplitudes in children ages 4.5–5.5 years, demonstrating the sensitivity of this neural measure of attention and inhibition to household income. In contrast, another study investigating ERP components putatively related to inhibitory control (N2) and attention (P3) – both aspects of executive function – in a group of 119 preschoolers (ages 4.5–5.6 years) found no such effects (Ruberry et al., 2016). Generally, both N2 and P3 components of the ERPs are elicited in the context of stimulus detection and implicated in selective attention. Although there were significant associations between income and behavioral performance on executive function tasks, no relationship between household income and ERP measures (N2 or P3) was detected (Ruberry et al., 2016).

To examine SES-related differences in functional brain development in preschoolers in Japan, Moriguchi and Shinohara (2019) measured prefrontal activation patterns during a cognitive set-shifting task using near-infrared spectroscopy (NIRS) in preschool children (ages 3.5–6.5 years). Activation in prefrontal regions of interest during performance on a card-sorting task (the Dimensional Change Card Sort – a putative measure of executive function) was correlated with SES measured via maternal education and household income. When a continuous composite SES score was used, no relationship with the prefrontal activation measures was observed. However, when SES was dichotomized into a categorical poverty vs. no poverty measure (with poverty defined as half the median household income of the total Japan's population), prefrontal activation (indexed with significant changes in oxy-hemoglobin in the task-switch vs. the baseline/rest conditions) was observed in the no-poverty group only, whereas the children experiencing poverty showed no significant activation during the task conditions (Moriguchi & Shinohara, 2019). Surprisingly, however, SES did not affect children's behavioral performance on the card-sorting task. Taken together, these findings of distinct neural profiles associated with low vs. high-SES children coupled with the lack of SES effects on task

performance may reflect compensatory mechanisms (e.g., recruiting other brain regions) adapted by these children in response to experiences associated with low SES, an interpretation consistent with other findings included in this review (e.g., Leonard et al., 2019).

Given known associations between SES and emerging language skills in early childhood (Hurt & Betancourt, 2017; Olson et al., 2020), there has been an interest in whether SES is associated with brain function underlying language processing in young children. Romeo et al. (2018) set out to examine how differences in home language exposure, such as the amount of words children heard at home and the number of conversational turns taken between children and their primary caregivers, may relate to brain function underlying language processing and to children's linguistic abilities, over and above the effects of SES measured with income and parental education. Using functional MRI to measure brain activation in language regions in response to speech, the authors found that 4.5–6.5 year-old children who experienced more conversational turns exhibited greater activation in the Broca's area during language processing, reflecting increased neural specialization for speech. Furthermore, together, conversational turns and Broca's area activation mediated the relationship between SES and children's language skills (Romeo et al., 2018).

In an earlier study, Raizada et al. (2008) examined whether five-year old children's language skills can be statistically predicted by the child's SES or brain activation in canonical language areas, including the inferior frontal gyrus (IFG). Using functional MRI, the authors measured brain activation during a rhyming task, which is predictive of reading skills. They observed that the rhyme task activation asymmetry (the difference between left and right IFG activity) was correlated with socioeconomic status measured with the Hollingshead index, reflecting increased specialization for language processing in higher-SES children, compared to lower-SES peers (Raizada et al., 2008).

Finally, SES-related differences in EEG patterns of brain activity have also been reported in a sample of 42 young children in Mexico (mean age ~4 years; Otero, 1997). Lower-SES children showed greater delta and theta absolute power in frontal leads than higher-SES children. Delta band oscillations are thought to relate to the synchronization of brain activity with autonomic function, motivational processes, and detecting saliency, whereas theta power putatively corresponds to states of drowsiness in children (Harmony, 2013). Lower-SES children also had reduced alpha power in occipital leads, compared to higher-SES peers. Together, these results raise the possibility of slower maturation of EEG-measured brain activity in lower-SES children, compared to their higher-SES peers (Otero, 1997).

Study Quality Assessment and Effect Sizes

Quality ratings of the reviewed studies – ranging from 2 to 6 on a 6-point scale of quality assessment – are shown in Table 2 and Table 3. Despite the substantial variability in study quality, all 19 studies included in the full text review stage were retained for the final qualitative synthesis. Thirteen of the 19 studies used multiple means of assessing SES (e.g., both parental education and income). Five of the studies included longitudinal analyses, allowing for investigations of SES links with brain developmental trajectories. Seventeen of the studies tested *a priori* hypotheses rather than conducting exploratory

analyses, providing increased power to detect effects and reducing the likelihood of committing a type I error. Cohorts included children from a wide array of racial, ethnic, and socioeconomic backgrounds, including young children from Japan, México, Spain, England, and many localities in the U.S. Although most studies (Betancourt et al., 2016; Gao et al., 2015; Hanson et al., 2013; Knickmeyer et al., 2017; Leonard et al., 2019; Qiu et al., 2017; Ruberry et al., 2016; Spann et al., 2020; Tomalski et al., 2013) included relevant covariates (e.g., gender, age, ethnicity), others neglected to control for these pertinent, possibly confounding variables, introducing error into their measurement, thereby limiting the conclusions that can be drawn from their results.

Effect sizes were extracted from 17 of the 19 studies (for two studies, data reported in the manuscript were insufficient to calculate effect size, and authors could not be reached or could not provide necessary data for effect size estimation). Effect sizes were estimated using either correlation coefficients (squared), t-statistics, or partial regression coefficients, and other necessary data (e.g., sample size, standard deviation of test statistics) using the “esc” package in R. Effect sizes (Hedges’ G) ranged from ~0 to 2.15. Effect size was not correlated with study quality ratings ($r = -0.04$, $p = 0.98$; see Figure 2).

Discussion

SES Associations with Brain Structure and Function: The present review included articles investigating associations between socioeconomic variables and neural measures in young children (≤ 5 years of age). Overall, findings indicate that correlations between various socioeconomic variables and brain structure and function can be detected at birth – suggesting that SES affects brain maturation in utero – and are identified throughout early childhood. This pattern of results addresses the first question of this review: are there associations between socioeconomic and neural measures in early childhood? The reviewed findings suggest that there are observable associations. Sixteen of the 19 studies reported SES associations with brain structure or function, and 17 of 19 also showed SES correlations with cognitive measures or developmental indices. While this pattern of results may reflect a “file drawer” bias (with null findings

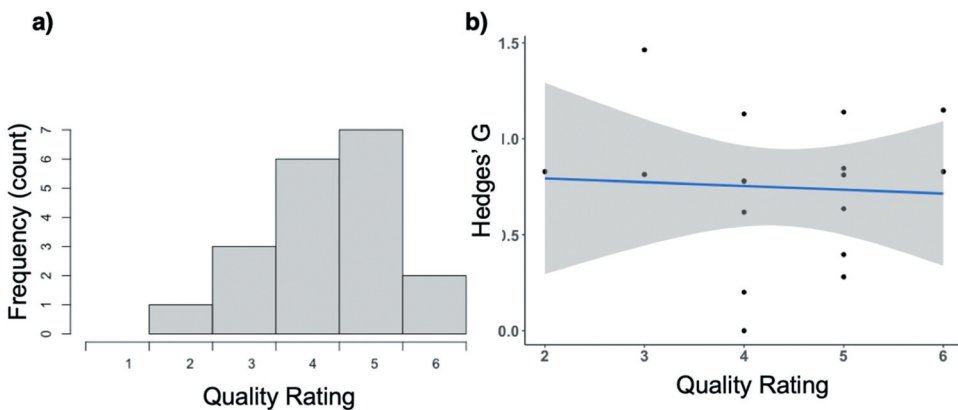


Figure 2. A) Histogram of quality ratings across studies, and B) scatter plot depicting the correlation between quality ratings and effect size (Hedges’ G), $r = -0.04$, $p = 0.98$.

more likely to be unpublished; Franco et al., 2014), the findings reviewed above provide compelling evidence for relationships between socioeconomic context and brain development in early childhood, prior to entry into kindergarten. Namely, the pattern of findings suggests that, broadly, socioeconomic context has wide-ranging implications for children's neurodevelopment during a sensitive period that sets the stage for their later ability to learn in school and beyond. These findings provide some neurobiological context for known socioeconomic achievement gaps that have been widely and consistently reported (Reardon et al., 2013).

Specificity of Associations: The results from the studies reviewed above also aid in addressing another question posed in this review: are SES associations with brain structure and function general or specific to certain regions and functions? Overall, results suggest that certain brain structures and functions may be more sensitive to socioeconomic contexts than others. In particular, regions and circuits supporting language processing and executive function appear to be particularly sensitive to socioeconomic context (e.g., Brooker, 2018; Conejero et al., 2018; Raizada et al., 2008; St. John et al., 2019). Namely, higher income is associated with higher cognitive flexibility and more mature neurocognitive network supporting executive function (as indexed with ERN component of the ERPs) in the second year of life (Conejero et al., 2018), and multifactorial SES is linked to brain activation in regions associated with language processing (with greater activation in the inferior frontal gyrus reflecting increased neural specialization for language; Raizada et al., 2008; Romeo et al., 2018).

The general pattern of findings emerging from the studies examining SES relationships with brain structure reveals reduced or decreased indices of cortical maturation in the first years of life (including lower cortical surface area, gray and white matter volume, and integrity of white matter tracts) in lower-SES children, compared to higher-SES peers. These differences in brain structure may point to alterations to developmental trajectories taking place very early in life, including before and shortly after birth (Betancourt et al., 2016; Hanson et al., 2013; Spann et al., 2020), which may reflect neurobiological processes directly affected by various SES-related corollaries (e.g., access to nutrition, prenatal care, stress, exposure to environmental pollutants and other teratogens). Although the mechanisms giving rise to the reported effects of low SES on brain structure are unknown and cannot be ethically studied in humans, evidence from animal studies using stress, maternal deprivation, or impoverished environment paradigms suggests that altered brain structure outcomes may be due to alterations in neuronal morphology and dendritic arborization (e.g., Eiland et al., 2012), stunted synaptogenesis (e.g., Liu et al., 2000), as well as suppression of neurogenesis (for reviews, see Hackman et al., 2010; Rutter, 2012), mediated through the hypothalamus-pituitary-adrenal (HPA) axis and epigenetic mechanisms.

Compensatory or adaptation mechanisms in response to stress, broadly defined and considered to be a major corollary of low SES, are also thought to mediate the effects of low SES on brain structural maturation through altering the duration of sensitive periods for certain aspects of structural brain development to take place, consistent with the stress acceleration hypothesis (Callaghan & Tottenham, 2016). One study reviewed here (Leonard et al., 2019) invoked these mechanisms, suggesting that a more extended, or less steep trajectory of cortical thinning observed in children from lower SES backgrounds may be an adaptive response to stress associated with socioeconomic disadvantage,

delaying maturation of certain brain circuits to allow the brain more time to respond to learning (Leonard et al., 2019).

Brain function (fMRI, EEG) associations with SES were also reported, in particular in regions associated with language and executive functions. These findings are consistent with the literature on robust links between SES and language development and learning in early childhood (Romeo et al., 2018). Findings on SES associations with fMRI measures were generally consistent across the age range studied, from infancy through early childhood (although Gao et al., 2015 reported only moderate SES-fMRI associations). However, as discussed in detail below (see *Future Directions*), given that most fMRI studies reviewed here used cross-sectional designs, longitudinal studies conducted on large samples would aid in determining whether the effects of SES on brain function strengthen over the course of development in early childhood.

There was greater variability in the findings on EEG measures of brain function as related to SES, compared to those using non-EEG methods. Of the seven studies using EEG or ERP indices brain function, five reported significant associations between SES and neural measures (Brooker, 2018; Conejero et al., 2018; Otero, 1997; St. John et al., 2019; Tomalski et al., 2013), with some contradictory results however, including reduced (Tomalski et al., 2013) or increased (Otero, 1997) EEG power observed in low-SES infants or preschoolers. The discrepancy could be either due to the developmental difference between infancy and later preschool age or relatively low study quality (see Table 3) suggesting a higher risk of biased results. There are several possible explanations for the lack of associations between SES and EEG/ERP measures (apart from a true lack of effect) reported in two out of seven reviewed studies (Brito et al.,; Ruberry et al., 2016), including suboptimal task demands used to elicit ERP components (limiting the neural responsivity to the task; Ruberry et al., 2016) or a very early age (12–96 hours after birth) for the SES-EEG links to manifest, suggesting that differences in postnatal experience may lead to the emergence of such links throughout early childhood (Brito et al.,).

Several ERP studies (Brooker, 2018; Conejero et al., 2018; St. John et al., 2019) focused on neurophysiological measures putatively related to executive control (e.g., the ERN, N2, and P3 components). The overall pattern of findings indicating associations between neural indices of executive control and SES suggests that some executive functions such as self-monitoring and inhibition may be particularly responsive to environmental context during early development.

Finally, only one study (Raizada et al., 2008) reported on findings using *both* functional and structural measures, making it difficult to assess whether brain structure or function may be more susceptible to SES-related alterations in brain development (although brain structure and function are inherently related; Gilmore et al., 2018). The primary hypotheses put forth by Raizada and colleagues addressed questions of brain function as related to SES, though their follow-up analyses on white and gray matter volume associations with SES were consistent with their functional findings, with both positively correlated with SES (Raizada et al., 2008).

Gradient vs. Dichotomous SES Variation: The third question addressed in this review relates to whether SES relations to brain development vary as a gradient, or whether there is a socioeconomic threshold under which brain development is impacted (e.g., poverty vs. no poverty). Most of the studies included in the review

(13 of 19) used continuous measures of SES, in alignment with recommendations for best practices of measuring SES (Diemer et al., 2013). Some employed both continuous and categorical measures, dichotomizing their SES variables into “low” and “high”; in one instance, this approach revealed that socioeconomic links with brain function were absent when using continuous measures, but detectable when SES was considered categorically (poverty vs. no poverty; Moriguchi & Shinohara, 2019). The majority of findings provide evidence for a socioeconomic gradient in measures of brain development (e.g., Raizada et al., 2008; Spann et al., 2020) and for continuous variability in neural measures associated with SES, although this may also reflect methods chosen by the researchers. In order to fully address whether SES-related variability in neural measures are continuous or discrete, studies would need to include analyses using multiple methods (see *Future Directions*). Additionally, dichotomizing SES into “high” vs. “low” is relative and is contingent on the sample- or context-specific measures (e.g., a U.S. federal poverty level may not be directly applicable as a poverty threshold in Spain, Japan, or Mexico, and vice versa). As such, inferences drawn from across studies need to be interpreted in the context of this limited generalizability.

Summary: Socioeconomic gaps in school and occupational achievement among school-age children and adults have been consistently documented. Early childhood is increasingly recognized as a time period that developmentally sets the stage for life-long cognitive skills and abilities (Gilmore et al., 2018). Findings from the present review suggest that, prior to entry into kindergarten, young children from communities and homes with reduced access to social and economic resources may already be at a disadvantage for learning, based on the observed differences in neural structure and function, with circuits subserving executive and language function being particularly sensitive to socioeconomic context (Moriguchi & Shinohara, 2019; Raizada et al., 2008; Romeo et al., 2018).

Limitations and Directions for Future Research: Most of the studies reviewed did not include young children with, or at risk for developing neurodevelopmental or psychiatric disorders. Two included children who were born pre-term, placing them at risk for a variety of neurodevelopmental delays, and one included children with heightened genetic risk for psychopathology (depressive disorder). This highlights a gap in the developmental psychopathology literature, as it remains unknown whether and to what extent low-SES may pose additional developmental risks for children with or at risk for developing neurodevelopmental or neuropsychiatric disorders such as autism, ADHD, anxiety, and depression. Given that there are socioeconomic health gradients across a variety of mental and physical health conditions, including some neurodevelopmental disorders (e.g., ADHD), investigations of whether and to what extent SES is associated with the development and severity of symptoms in psychological disorders are urgently needed.

Another factor limiting our interpretation across studies is the use of distinct measures to estimate SES, none without drawbacks. Perhaps the most debatable is the use of the original Hollingshead Index scale (Hollingshead, 1975), used by four of the studies reviewed (Brooker, 2018; Gao et al., 2015; Raizada et al., 2008; Spann et al., 2020). While the Hollingshead index combines multiple indices to estimate SES (e.g., education level, income, etc.), adding to the reliability of the measure (as compared to one index

alone), its “occupational prestige” scale is arguably outdated, reflecting societal biases rather than social or economic status per se (Duncan & Magnuson, 2012). Further, several studies dichotomized SES based on arbitrary thresholds of income or education (Betancourt et al., 2016; Moriguchi & Shinohara, 2019; Tomalski et al., 2013). However, there is no agreed-upon method for justifying a specific threshold for dichotomizing high vs. low-SES, and doing so may introduce further measurement error (Diemer et al., 2013).

All of the reviewed studies used socioeconomic variables that serve as proxies for other putatively causative variables, such as access to adequate nutrition, exposure to stress and environmental pollutants, access to physically, emotionally, and cognitively enriching environments, and ability for parents to spend quality time with their children (Farah, 2017). Thus, the observed patterns of findings do not reveal causal mechanisms underlying links between SES and brain structure and function pertaining to these underlying factors. However, they provide an important first step toward understanding *whether* there are SES relations with brain development before investigating what pathways lead to these associations (i.e., we must first understand the “what” in order to investigate “why”; Farah, 2017). The reviewed studies, including those with longitudinal design, do not address the issue of causation because low-SES is associated with multiple components of “risk,” and measured SES factors are likely only indicators of other critical underlying variables (which are increasingly being considered in recent years; e.g., neighborhood deprivation metrics).

Additionally, not all reviewed studies took into account, or covaried for factors that often confound SES, such as ethnicity and exposure to multiple languages in the home. Exposure to multiple languages may impact behavioral outcomes of language skills, as normative comparisons for these measures were monolingual, English-speaking children. Further, exposure to multiple language may be associated with brain function and structure (Abutalebi, Cappa & Perani, 2001; Mohades et al., 2012). Several studies did not mention, nor covary for biological sex, despite the known sex-dimorphic (neuroendocrine and immune) effects on early brain neurodevelopment (Arambula & McCarthy, 2020).

Despite these limitations, a major strength of this review is that it provides a first known systematic synthesis of the published research on the effects of SES on neurodevelopment in early childhood. To further understand how SES becomes embedded in the brain, early in life, future work needs to address *how* SES differences translate to alterations in brain structure and function using longitudinal studies with multiple socioeconomically related variables (e.g., characterizing nutrition status, exposure to environmental pollutants, home language and literacy environment, neighborhood-level characteristics, etc.) to understand causal relationships. Advanced analytical methods may be needed to address whether SES-related differences are continuous or operate on a threshold (i.e., using both dichotomous and continuous SES variables). Further, given the growing and compelling evidence for SES associations with *early* brain development reviewed here, a fruitful future direction for research would be inclusion of young children with or at risk for neurodevelopmental and/or neuropsychiatric conditions, to determine whether there may be multiplicative or additive risk associated with low-SES in these populations. Finally, and perhaps most importantly, studies aimed at addressing systemic causes of poverty are necessary and have the potential to influence

policy aimed at reducing societal inequity that creates biological, psychological, and cognitive disadvantage from the first years of life.

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